

# Kirlous Helal Zaki

.NET Backend Developer

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## CAREER OBJECTIVE

Motivated Computer Science graduate and aspiring .NET Backend Developer, with a strong foundation in RESTful API design, SQL Server, and clean architecture principles (MVVM, Repository Pattern). Experienced in building scalable backend systems and containerized deployments using Docker and Kubernetes. Skilled in Agile/Scrum environments and version control (Git). Reliable and detail-oriented, eager to contribute to a backend engineering team.

## EDUCATION

**Bachelor's Degree in Computer and Information Science,** Jul 2024  
*Ain Shams University*   
Scientific Computing Department Cumulative GPA 2.96/4.0 (Very Good)

## PROFESSIONAL EXPERIENCE

**AI Image Classification & Enhancement System (Python),** Jul 2024  
*Freelance collaboration with a PhD researcher*

- Trained and integrated multiple CNN/ML models into a single pipeline to predict digital-print quality factors (color type, contrast, resolution) from an image.
- Built a Tkinter GUI with image enhancement features (CMYK conversion, contrast, EDSR x3 upscaling) and packaged it as a standalone .exe

**Internship Automation and Manual API Testing, (Certificate)**  Jul 2024

- Gained experience in validating API functionality manually.
- Learned to automate test scripts for faster defect detection.

**MEAN Stack ITI Summer Code Camp, (Certificate)**  Aug 2022

- Developed front-end and back-end web applications using the MEAN stack.
- Integrated MEAN stack components and gained experience in building scalable web applications.

**Git/Github Summer Workshop** Sep 2022

- Applied Git and GitHub for version control, change tracking, and collaborative software development.

## COURSES

**Asp.Net Core Web Api**  Sep 2026

**Implementing and Developing SQL server objects**  Mar 2025

**Flutter Course Udemy**  Nov 2024

## CERTIFICATES

### Competitions

- Benha University Competition Certificate — Cardiac Function Assessment Graduation Project 
- Computer Vision Competition Certificate — Multi-Cancer Classification Project 
- Machine Learning Competition Certificate — Hotel Rating Prediction Project 

### Courses & Training

- Graduation from College of Computer Science, Ain Shams 
- Flutter Course Certificate 
- Internship Testing Certificate 
- MEAN Stack Development Course (ITI) 
- Military Service Certificate 

## PROJECTS

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### Cardiac Function Assessment(GP – Flutter & Deep Learning)

- **Problem Definition:** The left ventricle is crucial for pumping oxygenated blood, and its failure can lead to heart failure or cardiac arrest. Cardiomyopathy impacts its function, requiring early diagnosis, with echocardiograms assessing Left Ventricle Ejection Fraction (LVEF) as a key indicator of heart health.
- **Objective:** Developed a deep-learning system for echocardiogram analysis, achieving **91.8% DSC** in left-ventricle segmentation and **1.69 MSE**.
- **Backend & Integration:** Built a **FastAPI** backend to serve the trained model as a REST inference endpoint, and integrated it with a Flutter mobile app via HTTP requests for end-to-end video upload and real-time result delivery.
- **Competition:** Achieved **3rd Place** in Benha University Innovations Marathon. (**Competition Certificate**) 

### Cloud-Computing-Project (Docker & Kubernetes)

- Dockerized an existing e-commerce web application using **5 Docker images**, orchestrated via docker-compose and Kubernetes with **5 pods**, replica sets, and YAML deployment manifests.

### BuyingApp — E-Commerce Platform (MEAN Stack)

- **Team Project:** Built a full-stack e-commerce app (Angular, Node.js/Express, MongoDB) with **4+ core modules** — JWT authentication, home, product listing, and order management.

### Hotel Rating Prediction (Machine Learning)

- The project applies various machine learning algorithms for tasks like classification, regression, or clustering, focusing on building predictive models. It demonstrates hands-on problem-solving with real-world data, showcasing skills in model development and evaluation. (**Competition Certificate**) 

### Multi-Cancer Classification (Computer Vision)

- **Medical Image Classification:** Developed a computer vision system using machine learning and deep learning to classify medical images for multiple cancer types, applying image preprocessing and feature extraction techniques.
- **Model Performance:** Evaluated **5 models**, achieving **97.7% test accuracy** with Gradient Boosting, **96.5%** with AdaBoost, and **91.0%** with a custom CNN. (**Competition Certificate**) 

### Classical Encryption Techniques Package

- The project implements classical encryption techniques, including substitution ciphers (Caesar, Playfair, Polyalphabetic, Hill, Monoalphabetic) and transposition ciphers (Rail Fence, Columnar).

### Social App (Mobile App - Flutter)

- **Feature-Rich Social Platform:** Offers functionalities like user authentication, profile management, post sharing, and real-time interactions, creating an engaging social networking experience.
- **Firebase Integration:** Utilizes Firebase for backend services, including real-time database, authentication, and cloud storage, ensuring seamless and scalable performance.

## SKILLS

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### Technical Skills:

- ASP.NET Core Web API
- C#, C++, Java, Python, JavaScript
- SQL, SQL Server, NoSQL
- RESTful APIs
- Rest Assured (API Testing)
- Flutter, Dart (Mobile - secondary)
- Firebase, SQLite, Hive

### Soft Skills:

- Team Collaboration
- Analytical Skills
- Communication
- Leadership
- Problem Solving
- Work Independently
- Attention to Detail
- Innovation
- Adaptability
- Project Management
- Research
- Agile/Scrum

### Languages:

- Arabic – Native Proficiency
- English – Professional Working

### Tools & Platforms:

- Git & GitHub (Version Control)
- Visual Studio, Android Studio
- VS Code, Cursor, Windsurf
- Postman
- Docker
- Linux
- Katalon Studio
- Selenium
- Jupyter Notebook

### Core Concepts:

- .NET Core Backend
- MVVM Architecture
- Clean Architecture
- Repository Pattern
- API Design
- Database Management
- Object-Oriented Programming (OOP)
- Data Structures & Algorithms
- Testing Principles
- Network & Cybersecurity Fundamentals
- Angular Frontend
- AI/ML/Deep Learning